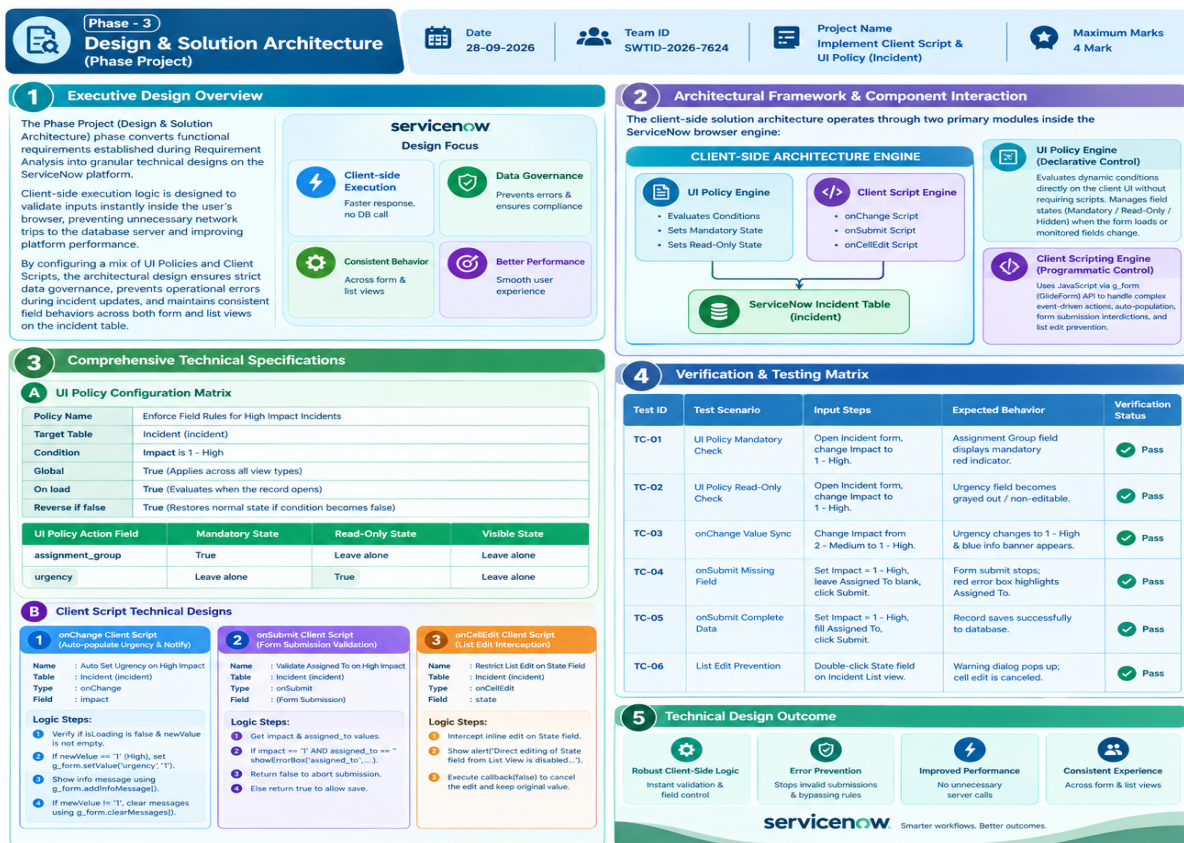


Phase-3 Design & Solution Architecture (Phase Project)

Field	Details
Date	28-09-2026
Team ID	SWTID-2026-8733
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Mark

Design & Solution Architecture

The Phase Project stage translates functional requirements into concrete platform configuration and technical script designs on the ServiceNow platform. This phase details the architectural framework, component interactions, field logic rules, and client-side execution steps required for the incident table



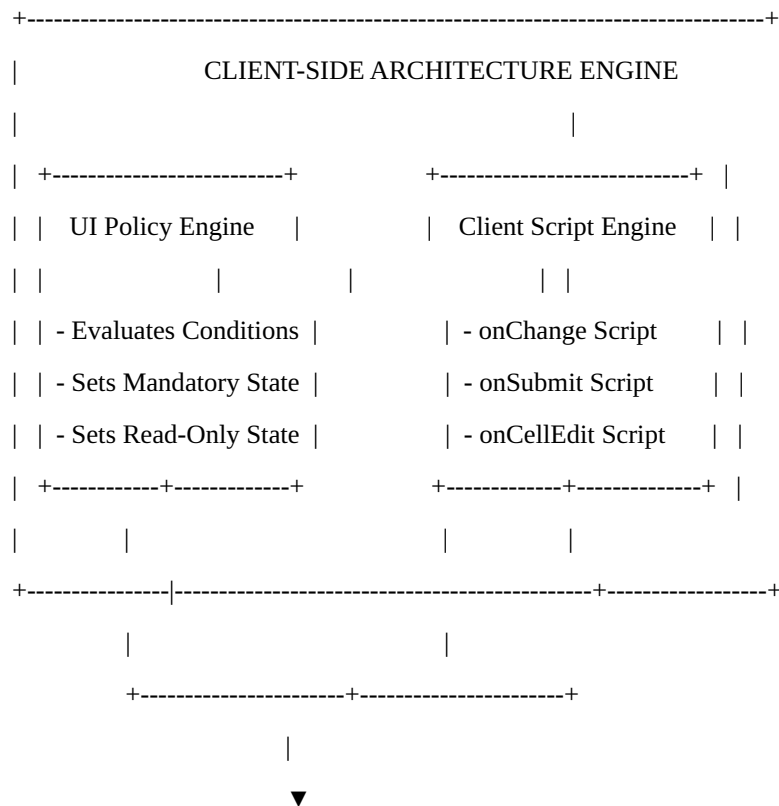
1. Executive Design Overview

The **Phase Project (Design & Solution Architecture)** phase converts functional requirements established during Requirement Analysis into granular technical designs on the ServiceNow platform. Client-side execution logic is designed to validate inputs instantly inside the user's browser, preventing unnecessary network trips to the database server and improving platform performance.

By configuring a mix of **UI Policies** and **Client Scripts**, the architectural design ensures strict data governance, prevents operational errors during incident updates, and maintains consistent field behaviors across both form and list views on the incident table.

2. Architectural Framework & Component Interaction

The client-side solution architecture operates through two primary modules inside the ServiceNow browser engine:



- 1. UI Policy Engine (Declarative Control):** Evaluates dynamic conditions directly on the client UI without requiring scripts. It actively manages field states (Mandatory / Read-Only / Hidden) whenever the form loads or monitored fields change.
- 2. Client Scripting Engine (Programmatic Control):** Uses JavaScript via the g_form (GlideForm) API to handle complex event-driven actions, auto-population, form submission interdictions, and list edit prevention.

3. Comprehensive Technical Specifications

A. UI Policy Configuration Matrix

- Policy Name:** Enforce Field Rules for High Impact Incidents
- Target Table:** Incident (incident)
- Condition:** Impact is 1 - High
- Global:** True (Applies across all view types)
- On load:** True (Evaluates when the record opens)

Reverse if false: True (Automatically restores normal state if condition becomes

UI Policy Action Field	Mandatory State	Read-Only State	Visible State
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assignment_group	True PDF	Leave alone	Leave alone
urgency	Leave alone	True	Leave alone

B. Client Script Technical Designs

1. onChange Client Script (Auto-populate Urgency & Notify)

- **Name:** Auto Set Urgency on High Impact
- **Table:** Incident (incident)
- **Type:** onChange
- **Field Name:** impact
- **Logic Execution Steps:**
 1. Verify if isLoading is false and newValue is not empty.
 2. If newValue == '1' (High Impact), invoke g_form.setValue('urgency', '1') to auto-populate Urgency to High.
 3. Trigger g_form.addInfoMessage('Urgency has been automatically updated to High due to High Impact choice.').
 4. If newValue != '1', clear former informational banners using g_form.clearMessages().

2. onSubmit Client Script (Form Submission Validation)

- **Name:** Validate Assigned To on High Impact Submit
- **Table:** Incident (incident)
- **Type:** onSubmit
- **Logic Execution Steps:**
 1. Retrieve impact value using g_form.getValue('impact').
 2. Retrieve assigned_to value using g_form.getValue('assigned_to').
 3. Evaluate condition: If impact == '1' AND assigned_to == " (empty):

- Execute `g_form.showErrorBox('assigned_to', 'Assigned To user is required before submitting a High Impact incident.')`.
- Return false to abort the save/submit process.

4. If condition fails, return true to allow the form to save successfully.

3. onCellEdit Client Script (List Edit Interception)

- **Name:** Restrict List Edit on State Field
- **Table:** Incident (incident)
- **Type:** onCellEdit
- **Field Name:** state
- **Logic Execution Steps:**
 1. Intercept user inline edit attempt on the State field from the list view.
 2. Prompt alert modal: `alert('Direct editing of State field from List View is disabled. Please open the Incident form to update State.')`.
 3. Execute `callback(false)` to discard changes and maintain original list value.
 4. **Verification & Testing Matrix**

Test ID	Test Scenario	Input Steps	Expected Behavior	Verification Status
TC-01	UI Policy Mandatory Check	Open Incident form, change Impact to 1 - High.	Assignment Group field displays mandatory red indicator	Pass
TC-02	UI Policy Read-Only Check	Open Incident form, change Impact to 1 - High.	Urgency field becomes grayed out / non-editable.	Pass
TC-03	onChange Value Sync	Change Impact from 2 - Medium to 1 - High. PDF	Change Impact from 2 - Medium to 1 - High. PDF	Pass
TC-04	onSubmit Missing Field	Set Impact = 1 - High, leave Assigned To blank, click Submit	Form submit stops; red error box highlights Assigned To.	Pass
TC-05	onSubmit Complete Data	Set Impact = 1 - High, fill Assigned	Record saves successfully to database. PDF	Pass

